Head & Neck Anatomy: TMJ & Muscles of Mastication

1. The temporomandibular joint (TMJ) is structurally classified as a specialized synovial joint. Morphologically, it belongs to which category?
a. Hinge joint
b. Pivot joint
c. Ball-and-socket joint
d. Plane joint
e. Condylar (or ginglymoarthrodial) joint
2. The upper articular surface of the TMJ is complex and non-flat. It is formed by the articular fossa (mandibular fossa) and the anteriorly situated convex:
a. Mastoid process
b. Styloid process
c. Articular eminence (tubercle) of the temporal bone
d. Infratemporal crest
e. Zygomatic process apex
3. The lower articular surface of the TMJ is formed by which dynamic component of the mandible?
a. Coronoid process
b. Angle of the mandible
c. Lingula
d. Head (condyle) of the mandible
e. Retromolar triangle
4. In a typical synovial joint, the articular surfaces are covered by hyaline cartilage. However, because the TMJ develops via intramembranous ossification, its articulating surfaces are uniquely covered by:
a. Yellow elastic cartilage
b. White fibrocartilage
c. Dense compact bone only
d. Synovial calcified sheets
e. Elastic connective bundles
5. The internal cavity of the TMJ is completely partitioned into an upper and a lower compartment by a specialized structure. This structure is the:
a. Fibrous capsule sleeve
b. Sphenomandibular ligament
c. Articular disc (composed of dense fibrocartilage)
d. Lateral ligament band
e. Stylomandibular ligament
6. The articular disc of the TMJ is concavo-convex on its superior surface and fits over the condyle inferiorly. It is divided anteroposteriorly into zones. The thin central area that lacks blood vessels and nerves is the:
a. Anterior extension
b. Anterior thick band
c. Posterior thick band
d. Intermediate zone
e. Bilaminar zone

7. The posterior region of the articular disc splits into upper and lower layers, wrapping around a highly vascularized, elastic neural tissue bed known as the:
a. Pterygoid fovea matrix
b. Retrodiscal tissue (bilaminar zone)
c. Sphenomandibular pad
d. Temporomandibular plexus
e. Submeniscal sinus
8. The fibrous capsule of the TMJ wraps completely around the joint. Superiorly, it attaches to the margins of the articular fossa; inferiorly, it attaches around the:
a. Coronoid process tip
b. Neck of the mandible
c. Angle of the jaw
d. Lingula crest
e. Mylohyoid ridge
9. The lateral aspect of the fibrous capsule is heavily reinforced by a true intrinsic joint ligament that stabilizes the condyle during jaw movements. This ligament is the:
a. Sphenomandibular ligament
b. Stylomandibular ligament
c. Lateral (temporomandibular) ligament
d. Pterygomandibular raphe
e. Stylohyoid ligament
10. The sphenomandibular ligament is an extrinsic joint stabilizer that runs from the spine of the sphenoid bone to the lingula of the mandible. Embryologically, it represents a remnant of:
a. Reichert's cartilage
b. Meckel's cartilage (1st arch)
c. Third arch skeletal core
d. The styloid process blastema
e. The primitive tympanic ring
11. The stylomandibular ligament is a specialized band of deep fascia stretching from the styloid process to the posterior border of the mandibular angle. Morphologically, it is a thickening of the:
a. Deep cervical fascia investing layer
b. Parotid fascia
c. Lamina papyracea
d. Deep cervical fascia parapharyngeal layer
e. Temporal fascia
12. The structural barrier that limits excessive retraction or backward displacement of the mandibular condyle, protecting the external acoustic meatus, is the:
a. Sphenomandibular ligament
b. Lateral (temporomandibular) ligament
c. Stylomandibular ligament
d. Mylohyoid fascia sheet
e. Pterygomandibular raphe
13. The TMJ is capable of executing two types of kinematics. Gliding or translational movements take place exclusively within which joint space?
a. Lower compartment (inframeniscal space)
b. Pterygoid fovea depression

c. Upper compartment (suprameniscal space)
d. Mandibular canal tract
e. Retrodiscal fat pad

14. Hinge-like rotational movements of the mandible, which occur during the initial phases of jaw opening, take place entirely within the:

a. Upper compartment (suprameniscal space)
b. Lower compartment (inframeniscal space)
c. Infratemporal roof matrix
d. Sublingual salivary space
e. Retromolar triangle plane

15. Muscles of mastication drive all mandibular movements. The temporalis muscle is a fan-shaped muscle that inserts onto the coronoid process and acts primarily to:

a. Depress and protrude the mandible
b. Elevate and retract (posterior fibers) the mandible
c. Abduct and rotate the jaw laterally
d. Depress the hyoid bone
e. Compress the oral vestibule walls

16. The masseter muscle is a thick, quadrilateral muscle covering the lateral surface of the ramus. Its primary mechanical action is to:

a. Depress the mandible to open the mouth
b. Retract the condyle into the fossa
c. Elevate the mandible to close the mouth powerfully
d. Rotate the jaw into a wide lateral smile
e. Draw the tongue backward during swallowing

17. The medial pterygoid muscle forms a structural muscular sling with the masseter around the angle of the jaw. When both medial pterygoid muscles contract simultaneously, they cause:

a. Mandibular depression
b. Mandibular retraction
c. Mandibular elevation and protrusion
d. Selective condylar torsion
e. True rotational tilt of the TMJ axis

18. The lateral pterygoid muscle consists of two anatomical heads. The inferior head originates from the lateral surface of the lateral pterygoid plate and inserts directly into the:

a. Coronoid process medial wall
b. Pterygomandibular raphe core
c. Pterygoid fovea on the neck of the mandible
d. Angle of the jaw inner aspect
e. Mylohyoid groove floor

19. The superior head of the lateral pterygoid muscle originates from the infratemporal surface of the greater wing of the sphenoid and inserts into the:

a. Mandibular angle edge
b. Capsule and anterior margin of the TMJ articular disc
c. Retromolar triangle fossa
d. Stylomandibular ligament trunk
e. Temporal crest line

20. Which muscle of mastication is functionally unique because it acts as the primary initializer and executor of jaw depression (opening the mouth)?
a. Masseter muscle
b. Medial pterygoid muscle
c. Temporalis muscle
d. Lateral pterygoid muscle
e. Buccinator muscle
21. Side-to-side chewing or grinding movements of the jaw are produced by the alternate, coordinated contraction of which pair of muscles?
a. Right and left temporalis muscles
b. Right and left masseter muscles
c. Medial and lateral pterygoid muscles of opposite sides acting sequentially
d. Orbicularis oris and buccinator muscles
e. Anterior and posterior bellies of the digastric
22. All four principal muscles of mastication develop from the mesenchyme of the same embryonic structure. This source is the:
a. Frontonasal prominence
b. First pharyngeal (mandibular) arch
c. Second pharyngeal (hyoid) arch
d. Third pharyngeal pouch
e. Occipital somites row
23. Because of their shared pharyngeal arch origin, all muscles of mastication receive their motor innervation from branches of which nerve trunk?
a. Maxillary nerve (V2)
b. Ophthalmic nerve (V1)
c. Mandibular nerve (V3 division of Trigeminal)
d. Facial nerve (CN VII)
e. Glossopharyngeal nerve (CN IX)
24. The sensory innervation to the capsule and structures of the TMJ is mediated primarily by two branches of the mandibular nerve. These branches are the:
a. Long buccal and mental nerves
b. Deep temporal and masseteric nerves
c. Auriculotemporal and masseteric nerves
d. Lingual and inferior alveolar nerves
e. Incisive and mylohyoid nerves
25. The auriculotemporal nerve emerges from the infratemporal fossa and winds upward immediately behind the TMJ capsule. This close anatomical relationship makes it highly vulnerable to compression during:
a. Dislocation or inflammatory conditions of the condyle
b. Fractures of the symphysis menti
c. Surgical extractions of upper incisors
d. Anesthesia blocks of the infraorbital nerve
e. Hyperplasia of the mental spines
26. The high-pressure arterial blood supply to the TMJ is provided by branches of the external carotid artery. The two primary vessels supplying the joint are the:
a. Facial and lingual arteries
b. Maxillary and superficial temporal arteries
c. Internal carotid and middle meningeal arteries

d. Infraorbital and superior alveolar arteries
e. Occipital and posterior auricular arteries

27. Venous blood from the TMJ capillary beds drains backward to empty directly into which extensive vascular network located in the infratemporal fossa?

a. Cavernous sinus
b. Pterygoid venous plexus
c. Pharyngeal venous plexus
d. Internal jugular vein
e. Submental plexus

28. Dislocation of the TMJ occurs when a patient opens their mouth excessively wide (such as during a massive yawn or extraction of a lower molar), causing the condyle to slide out of its socket. The condyle slips over the:

a. Postglenoid tubercle posteriorly into the ear canal
b. Articular eminence anteriorly into the infratemporal fossa
c. Greater wing of sphenoid superiorly into the brain cavity
d. Styloid process medially
e. Zygomatic bone laterally through the skin

29. When the TMJ is dislocated anteriorly, the patient presents clinically with a locked jaw. The jaw remains fixed in an open position because the masseter, temporalis, and medial pterygoid muscles undergo:

a. Flaccid paralysis
b. Chronic wasting and atrophy
c. Protective spasm (reflex trismus) that holds the condyle trapped ahead of the eminence
d. Complete structural rupture
e. Avascular necrosis

30. To reduce an anterior TMJ dislocation extraorally, the clinician places their thumbs on the lower molar teeth and applies steady mechanical force directed:

a. Downward and then backward
b. Purely upward and forward
c. Laterally and then superiorly
d. Medially into the palate box
e. Straight forward into the nose axis

31. Temporomandibular joint disorder (TMD or TMJ syndrome) often manifests as jaw pain, clicking sounds during chewing, and restricted movement. Pain from the joint can radiate to the temple or ear because they share sensory supply via the:

a. Facial nerve
b. Auriculotemporal nerve
c. Long buccal nerve
d. Mental nerve
e. Great auricular nerve

32. The sphenomandibular ligament lies medial to the TMJ capsule. The space situated between this ligament and the neck of the mandible transmits which high-yield vascular and neural pair?

a. Lingual nerve and facial artery
b. Maxillary artery and inferior alveolar nerve
c. Superficial temporal artery and facial nerve trunk
d. Middle meningeal artery and auriculotemporal nerve roots
e. Deep temporal artery and long buccal nerve

33. The masseteric nerve passes from the infratemporal fossa to reach the deep surface of the masseter muscle by crossing through which bony landmark?
a. Foramen ovale
b. Foramen spinosum
c. Mandibular notch (incisura)
d. Pterygomaxillary fissure
e. Inferior orbital fissure
34. The masseteric nerve is a direct branch of which division of the mandibular nerve (V3)?
a. Main trunk before splitting
b. Anterior division (primarily motor)
c. Posterior division (primarily sensory)
d. Otic ganglion branch pack
e. Nervus spinosus loop
35. The deep temporal nerves supply the powerful temporalis muscle. These nerves run superiorly from the infratemporal fossa, passing deep to the:
a. Zygomatic arch and temporalis tendon
b. Condylar process of the mandible
c. Lateral pterygoid plate surface
d. Pterygoid hamulus bone curve
e. Floor of the orbit plate
36. The nerve that supplies the medial pterygoid muscle arises directly from the main trunk of V3. This nerve is unique because it passes through or close to which peripheral parasympathetic ganglion without synapsing?
a. Ciliary ganglion
b. Submandibular ganglion
c. Pterygopalatine ganglion
d. Otic ganglion
e. Geniculate ganglion
37. The nerve to the medial pterygoid also provides motor supply to two separate small skeletal muscles. These are the:
a. Buccinator and risorius
b. Tensor veli palatini and tensor tympani muscles
c. Anterior and posterior bellies of digastric
d. Mylohyoid and geniohyoid muscles
e. Stylohyoid and stylopharyngeus muscles
38. The lateral pterygoid nerve supplies the lateral pterygoid muscle. It arises as a direct motor branch from the:
a. Posterior division of V3
b. Anterior division of V3
c. Long buccal nerve trunk purely
d. Mental nerve bifurcation
e. Facial nerve zygomatic loop
39. Protrusion (forward translation) of the mandible requires the bilateral, simultaneous contraction of which muscle group?
a. Masseter and temporalis muscles
b. Lateral and medial pterygoid muscles acting together
c. Right and left anterior bellies of digastric

d. Platysma sheets
e. Orbicularis oris sphincter loops

40. Retraction (backward translation) of the mandible, pulling the condyles back into the mandibular fossae from a protruded position, is driven primarily by the:
a. Inferior head of the lateral pterygoid muscle
b. Superficial head of the medial pterygoid muscle
c. Posterior horizontal fibers of the temporalis muscle
d. Anterior vertical fibers of the masseter muscle
e. Buccinator fibers

41. Severe, painful spasm of the muscles of mastication that restricts or completely prevents a patient from opening their mouth is a critical clinical condition known as:
a. Epiphora
b. Trismus (lockjaw)
c. Sialorrhea
d. Ptosis
e. Cycloplegia

42. Trismus can occur due to a variety of dental causes, including a localized spread of infection into the infratemporal or pterygomandibular space from an impacted lower:
a. Central incisor
b. Canine
c. First premolar
d. Third molar (wisdom tooth)
e. Maxillary lateral incisor

43. The pterygomandibular raphe is a vital landmark in local anesthesia, separating the buccinator muscle from the superior pharyngeal constrictor. It attaches superiorly to the pterygoid hamulus and inferiorly to the:
a. Angle of the mandible
b. Mylohyoid line of the mandible (posterior end)
c. Coronoid process tip
d. Condylar neck fovea
e. Mental tubercle line

44. To perform an inferior alveolar nerve block safely, the dental needle targets the pterygomandibular space. This anatomical space is bounded laterally by the ramus of the mandible and medially by the:
a. Masseter muscle
b. Medial pterygoid muscle
c. Lateral pterygoid muscle
d. Temporalis muscle insert
e. Superior pharyngeal constrictor

45. If the dental needle is accidentally redirected too far posteriorly during an inferior alveolar nerve block, it will pierce the fascial capsule of the parotid gland. Depositing anesthetic solution here will anesthetize which nerve?
a. Trigeminal nerve main trunk
b. Lingual nerve
c. Facial nerve (CN VII terminal motor branches)
d. Auriculotemporal nerve
e. Glossopharyngeal nerve

46. Accidental anesthesia of the facial nerve within the parotid gland capsule leads to a transient, reversible complication characterized by:
a. Inability to sense taste on the tongue
b. Severe numbness over the chin skin
c. Transient facial paralysis (inability to close the eye or wrinkle the forehead on that side)
d. Acute dry mouth due to parotid shutdown
e. Absolute lockjaw trismus
47. The sensory nerve that exits the infratemporal fossa by passing between the two heads of the lateral pterygoid muscle to supply the skin and mucosa of the cheek is the:
a. Buccal branch of the facial nerve
b. Long buccal nerve (branch of V1/V3)
c. Inferior alveolar nerve
d. Mental nerve trunk
e. Auriculotemporal nerve
48. Structural grinding of teeth or clenching of the jaw during sleep, known clinically as bruxism, can overwork the muscles of mastication and lead to secondary hypertrophy of the:
a. Digastric muscle
b. Mylohyoid muscle
c. Masseter muscle (causing a widened jawline configuration)
d. Platysma sheets
e. Tensor veli palatini muscle
49. The capsule of the TMJ is unique because it is reinforced by the lateral ligament, whose fibers run obliquely downward and backward. This specific orientation works mechanically to prevent:
a. Excessive depression of the jaw
b. Posterior and lateral displacement of the condyle
c. Anterior translation during normal speech
d. Synovial fluid filtration out of the upper space
e. Elongation of the coronoid process
50. During extraoral surgical approaches to access a fractured condylar neck, a dental or maxillofacial surgeon must be extremely careful when retracting the tissue behind the ramus to avoid severing which vertical blood vessel?
a. Maxillary artery main trunk
b. Superficial temporal artery
c. Facial artery
d. Middle meningeal artery
e. Lingual artery

Answer Key Table (TMJ & Mastication - Sequence 3 Loop)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50
e	**c**	**c**	**a**	**c**	**b**	**b**

c	**b**	**c**	**b**	**b**	**d**	**c**
d	**a**	**b**	**b**	**a**	**b**	**b**
c	**b**	**b**	**b**	**d**	**b**	**a**
c	**b**	**b**	**c**	**b**	**c**	
d	**c**	**c**	**c**	**b**	**c**	
b	**c**	**b**	**b**	**c**	**b**	
b	**c**	**c**	**a**	**a**	**c**	